

# USE CASE: LOG GRANT EXPENSE

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**Actor:** Faculty Researcher

**Preconditions:**

- Faculty Researcher is authenticated and logged into the system.
- Faculty Researcher has at least one active, approved grant assigned to them.

**Postconditions:**

- The expense is recorded against the grant's ledger.
- The grant's remaining budget balance is updated to reflect the new expense.
- An immutable audit log entry is created for the transaction (as per NFR-001).

**Main Success Scenario:**

1. Faculty Researcher selects an active grant from their dashboard.
2. Faculty Researcher selects "Log Expense" and enters expense details.
3. The system validates the input fields for completeness and correct format.
4. The system checks the expense amount against the grant's remaining budget balance (include: Check Budget Balance).
5. The system confirms the expense does not exceed the remaining balance.
6. The system records the expense, updates the running budget balance, and writes an audit log entry.

7. The system displays a confirmation with the updated balance to the Faculty Researcher.

**Alternate Flow:**

- 4a. At step 5, the system detects the expense would cause the balance to go negative.
- 4b. The system blocks the transaction and displays an error indicating the expense exceeds the remaining budget.
- 4c. The system notifies the Research Dean that an override may be required.
- 4d. If the Research Dean approves an override, the system re-attempts step 6 with the override flag set and logs the override action separately in the audit trail.
- 4e. If no override is granted, the expense remains unrecorded and the Faculty Researcher is returned to step 2.